

ASSESSMENT TOOL 3

ITA0609 - Machine Learning

Sameer Ahmed G
192421154

Question 1 Backpropagation in a Multilayer Perceptron for Binary Classification

Introduction:

Artificial Neural Networks (ANNs) are one of the most widely used machine learning models for solving classification, prediction, and pattern recognition problems. A **Multilayer Perceptron (MLP)** is a type of feed-forward neural network that consists of an input layer, one or more hidden layers, and an output layer. Unlike a single-layer perceptron, an MLP can solve complex and non-linearly separable problems by learning hidden patterns from the data. The learning process of an MLP is performed using the **Backpropagation Algorithm**, which updates the weights of the network by minimizing the prediction error. During training, the network repeatedly compares its predicted output with the actual output and adjusts its weights to improve accuracy. Backpropagation is therefore considered one of the most important learning algorithms in deep learning and artificial intelligence.

In binary classification problems, the objective of the neural network is to classify an input into one of two classes, such as **Yes/No**, **Disease/No Disease**, **Spam/Not Spam**, or **Approved/Rejected**. The network learns from labelled examples by reducing the difference between the predicted output and the expected output. This reduction in error is achieved through forward propagation followed by backward propagation of the error.

Architecture of a Multilayer Perceptron

A Multilayer Perceptron consists of three main layers:

1. Input Layer

The input layer receives the feature values from the dataset. Each neuron represents one input attribute. For example, in a medical diagnosis system, the input neurons may represent Age, Blood Pressure, Glucose Level, and Lifestyle Risk.

2. Hidden Layer

The hidden layer performs mathematical computations and extracts meaningful patterns from the input data. Multiple hidden layers enable the network to learn complex relationships between different features.

3. Output Layer

The output layer produces the final prediction. In binary classification, the output neuron usually returns either **1 (Yes)** or **0 (No)** after applying an activation function.

Working of Backpropagation

Backpropagation is a supervised learning algorithm that updates the connection weights by propagating the prediction error backward through the network. The algorithm consists of three major phases:

- Forward Pass
- Error Calculation
- Backward Propagation and Weight Update

These steps are repeated until the network achieves minimum prediction error.

Step 1: Forward Pass

During the forward pass, the input values are multiplied by their corresponding weights, added together with a bias value, and passed through an activation function. The output of one layer becomes the input to the next layer until the final prediction is obtained.

The mathematical equation for the net input is

$$z = \sum_{i=1}^n x_i w_i + b$$

where

- x_i = Input values
- w_i = Weights
- b = Bias

The output is then calculated using an activation function.

For binary classification, the **Sigmoid Activation Function** is commonly used.

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

The sigmoid function converts any real value into a probability between 0 and 1.

Simple Example

Consider a neural network with two input features.

Inputs

$$\begin{aligned} x_1 &= 1 \\ x_2 &= 0 \end{aligned}$$

Weights

$$w_1 = 0.4$$

$$w_2 = 0.6$$

Bias

$$b = 0.2$$

Calculate Net Input

$$\begin{aligned} z &= (1 \times 0.4) + (0 \times 0.6) + 0.2 \\ &= 0.4 + 0 + 0.2 \\ &= 0.6 \end{aligned}$$

Now apply the sigmoid activation function.

$$\begin{aligned} \sigma(0.6) &= \frac{1}{1 + e^{-0.6}} \\ \sigma(0.6) &\approx 0.646 \end{aligned}$$

Therefore,

Predicted Output = 0.646

This means the network predicts approximately **64.6% probability** that the sample belongs to the positive class.

Step 2: Error Calculation

After obtaining the predicted output, it is compared with the actual target value.

Suppose

Actual Output

$$y = 1$$

Predicted Output

$$\hat{y} = 0.646$$

The prediction error is calculated using the Mean Squared Error (MSE).

$$Error = \frac{1}{2} (y - \hat{y})^2$$

Substitute the values.

$$\begin{aligned} &= \frac{1}{2}(1 - 0.646)^2 \\ &= \frac{1}{2}(0.354)^2 \\ &= \frac{1}{2}(0.125) \\ &= 0.0625 \end{aligned}$$

The error indicates how far the prediction is from the expected output. The objective of training is to reduce this error as much as possible.

Step 3: Backward Propagation

During backward propagation, the error is propagated from the output layer towards the hidden layer. The contribution of each weight to the total error is calculated using the chain rule of differentiation. Based on this contribution, each weight is adjusted to reduce future prediction errors.

The general weight update formula is

$$\text{New Weight} = \text{Old Weight} - \eta \frac{\partial E}{\partial W}$$

where

- E = Error
- W = Weight
- η = Learning Rate

Assume

Learning Rate

$$\eta = 0.1$$

Suppose the calculated gradient for weight w_1 is

$$0.25$$

Then

$$\begin{aligned} \text{New Weight} &= 0.4 - (0.1 \times 0.25) \\ &= 0.4 - 0.025 \\ &= 0.375 \end{aligned}$$

Similarly, all other weights are updated.

After weight updates, another forward pass is performed.

This process continues for multiple epochs until the network converges to the minimum error.

Flow of Backpropagation

Input Data

|



Initialize Weights

|



Forward Pass

|



Predicted Output

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Calculate Error

|



Backward Propagation

|



Update Weights

|



Repeat Until Error is Minimum

Role of Learning Rate

The learning rate determines the size of each weight update during training. It controls how quickly the neural network learns from the training data.

If the learning rate is **too small**, the network learns very slowly and requires many iterations to converge.

If the learning rate is **too large**, the network may overshoot the minimum error, causing unstable learning and poor prediction accuracy.

A properly selected learning rate ensures faster convergence and stable optimization.

Typical learning rate values are:

- 0.001
- 0.01
- 0.1

Adaptive optimizers such as **Adam**, **RMSProp**, and **Adagrad** automatically adjust the learning rate during training for better performance.

Role of Activation Functions

Activation functions introduce non-linearity into the neural network, enabling it to learn complex relationships between input features and output classes. Without activation functions, a neural network would behave like a simple linear model regardless of the number of hidden layers.

Common activation functions include:

Sigmoid Function

Range: 0 to 1

Advantages:

- Suitable for binary classification.
- Produces probability values.
- Easy to interpret.

Disadvantages:

- Suffers from vanishing gradient problem.

ReLU (Rectified Linear Unit)

$$f(x) = \max(0, x)$$

Advantages:

- Faster training.
- Reduces vanishing gradients.
- Widely used in hidden layers.
-

Tanh Function

Range:

-1 to +1

Advantages:

- Zero-centered output.
- Better gradient flow than sigmoid.

Advantages of Backpropagation

Backpropagation offers several important advantages:

- Automatically updates network weights.
- Minimizes prediction error efficiently.
- Learns complex nonlinear relationships.
- Supports multiple hidden layers.
- Provides high classification accuracy.
- Forms the basis of modern deep learning models.

Applications

Backpropagation is widely used in various real-world applications, including:

- Medical Diagnosis
- Disease Prediction
- Face Recognition
- Speech Recognition
- Handwriting Recognition
- Autonomous Vehicles
- Image Classification
- Fraud Detection
- Sentiment Analysis
- Recommendation Systems

Conclusion

Backpropagation is the fundamental learning algorithm used in Multilayer Perceptrons for binary classification. During training, the algorithm performs a forward pass to compute predictions, calculates the prediction error, and propagates this error backward through the network to update the weights. The learning rate controls the speed of weight updates, while activation functions enable the network to model complex nonlinear relationships. By repeatedly reducing prediction error over multiple iterations, backpropagation allows neural networks to achieve high accuracy and reliable performance across a wide range of machine learning applications.

Question 2 Genetic Algorithm for Optimizing Candidate Solutions

Introduction

A **Genetic Algorithm (GA)** is a population-based optimization technique inspired by Charles Darwin's theory of natural evolution. It searches for the optimal solution by repeatedly selecting the fittest individuals and applying genetic operators such as **selection, crossover, and mutation**. Instead of evaluating a single solution at a time, a Genetic Algorithm works with a population of candidate solutions and gradually improves them over several generations. This evolutionary process enables the algorithm to efficiently solve complex optimization problems where traditional methods may fail. Genetic Algorithms are widely used in machine learning, engineering design, robotics, scheduling, feature selection, and artificial intelligence because of their ability to explore large search spaces and avoid getting trapped in local optimum solutions.

In this problem, a dataset representing different candidate solutions based on **Speed** and **Accuracy** is considered. The objective is to maximize the fitness function by selecting the best individuals and evolving them through successive generations until the optimal solution is obtained.

Initial Dataset

Candidate	Speed	Accuracy	Suitable
A	70	75	Yes
B	60	65	No
C	85	80	Yes
D	90	95	Yes

For simplicity, the fitness function is defined as the sum of Speed and Accuracy.

$$Fitness = Speed + Accuracy$$

Step 1: Fitness Evaluation

The first step of the Genetic Algorithm is to calculate the fitness value for every candidate solution.

For Candidate A

$$Fitness = 70 + 75 = 145$$

For Candidate B

$$Fitness = 60 + 65 = 125$$

For Candidate C

$$Fitness = 85 + 80 = 165$$

For Candidate D

$$Fitness = 90 + 95 = 185$$

The calculated fitness values are shown below.

Candidate	Fitness
A	145
B	125
C	165
D	185

The higher the fitness value, the better the candidate solution. Candidate D has the highest fitness, followed by Candidate C.

Step 2: Selection

Selection is the process of choosing the fittest individuals from the current population for reproduction. Candidates with higher fitness values have a greater probability of passing their characteristics to the next generation. Various selection methods such as Roulette Wheel Selection, Tournament Selection, and Rank Selection can be used. In this example, the two candidates with the highest fitness values are selected.

Selected Parents:

Parent 1 = Candidate D

Speed = 90

Accuracy = 95

Fitness = 185

Parent 2 = Candidate C

Speed = 85

Accuracy = 80

Fitness = 165

These individuals are selected because they possess the best characteristics among the available population.

Step 3: Binary Representation

Before performing crossover, represent the Speed values in binary form.

90

1011010

85

1010101

Step 4: Crossover

Crossover combines the genetic information of two parents to generate new offspring. A single-point crossover is performed after the fourth bit.

Parent 1

1011 | 010

Parent 2

1010 | 101

After exchanging the remaining bits,

Offspring 1

1011 | 101

Binary = 93

Offspring 2

1010 | 010

Binary = 82

Assume the Accuracy values are also exchanged.

Offspring 1

Speed = 93

Accuracy = 95

Fitness

$$93 + 95 = 188$$

Offspring 2

Speed = 82

Accuracy = 80

Fitness

$$82 + 80 = 162$$

The new population becomes

Candidate	Speed	Accuracy	Fitness
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Offspring 1	93	95	188
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Offspring 2	82	80	162
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The first offspring now has a higher fitness than either parent, indicating an improvement in the population.

Step 5: Mutation

Mutation randomly changes one or more genes of a chromosome after crossover. Mutation prevents the algorithm from becoming trapped in local optimum solutions and maintains genetic diversity within the population.

Suppose the binary chromosome of Offspring 1 is

1011101

Mutate the last bit

1011111

Binary value becomes

95

The new candidate is

Speed = 95

Accuracy = 95

Fitness

$$95 + 95 = 190$$

The updated population is

Candidate	Speed	Accuracy	Fitness
Mutated Offspring	95	95	190

The mutation operation has successfully produced a better solution than the previous generation.

Evolution Across Generations

Generation	Best Candidate	Fitness
Generation 1	D	185
Generation 2	Offspring 1	188
Generation 3	Mutated Offspring	190

The fitness value increases from **185** to **188**, and finally to **190**, demonstrating continuous improvement through evolutionary operations.

Role of Selection

Selection is responsible for choosing the fittest individuals for reproduction. Strong candidates have a higher probability of producing offspring, ensuring that desirable characteristics are preserved in future generations. Selection accelerates convergence toward the optimal solution while eliminating weak individuals from the population.

Role of Crossover

Crossover combines the characteristics of two parent chromosomes to produce new offspring. It creates new candidate solutions by exchanging genetic information, allowing the algorithm to explore better combinations of parameters. Crossover increases the search capability of the Genetic Algorithm and helps discover higher-quality solutions.

Role of Mutation

Mutation introduces random modifications into chromosomes after crossover. This operation maintains diversity within the population, prevents premature convergence, and enables the algorithm to escape local optimum solutions. Mutation ensures continuous exploration of the search space, increasing the probability of finding the global optimum.

Applications of Genetic Algorithms

Genetic Algorithms are extensively applied in machine learning, feature selection, neural network optimization, scheduling, robotics, engineering design, image processing, path planning, financial optimization, cloud computing, and data mining. Their ability to efficiently search complex solution spaces makes them suitable for solving large-scale optimization problems.

Advantages

Genetic Algorithms can handle nonlinear optimization problems, search large solution spaces efficiently, avoid local optimum solutions, require no gradient information, and provide highly optimized solutions for complex real-world applications.

Conclusion

The Genetic Algorithm successfully improves the quality of candidate solutions through repeated application of **selection, crossover, and mutation**. Selection chooses the fittest individuals, crossover combines their characteristics to generate better offspring, and mutation introduces diversity that prevents premature convergence. Starting with an initial best fitness of **185**, the evolutionary process increases the fitness to **188** after crossover and finally to **190** after mutation. Therefore, the **final optimized solution** is **Speed = 95, Accuracy = 95, Fitness = 190**, demonstrating how Genetic Algorithms efficiently evolve toward the best possible solution.

Question 3 Neural Network Classification Using Backpropagation and Model Evaluation

Introduction

Artificial Neural Networks (ANNs) are powerful supervised machine learning models that imitate the working of the human brain to solve classification and prediction problems. A neural network consists of interconnected neurons arranged in an input layer, one or more hidden layers, and an output layer. During training, the network learns the relationship between input features and output classes by adjusting its connection weights using the **Backpropagation Algorithm**. Backpropagation minimizes prediction error by propagating the error from the output layer to the hidden layers and updating the weights iteratively. Neural networks are widely used in healthcare for disease diagnosis because they can identify complex patterns that traditional algorithms may fail to recognize. In this problem, a neural network is trained to predict whether a patient has a disease based on the attributes **Age, Blood Pressure, Glucose Level, and Lifestyle Risk**. After training, the performance of the model is evaluated using a confusion matrix and several classification metrics.

Given Dataset

Example	Age	Blood Pressure	Glucose Level	Lifestyle Risk	Actual Output	Predicted Output
1	45	High	High	High	Yes	Yes
2	50	Normal	High	Medium	Yes	No
3	30	Normal	Normal	Low	No	No
4	60	High	High	High	Yes	Yes
5	40	Normal	Normal	Medium	No	Yes

The objective is to train a neural network using this dataset and evaluate its classification performance by comparing the predicted outputs with the actual outputs.

Training the Neural Network

The neural network training process begins by receiving the input attributes from the dataset. Each input feature is connected to neurons in the hidden layer through weighted connections. Initially, these weights are assigned random values. During the forward pass, each neuron computes a weighted sum of its inputs, adds a bias value, and applies an activation function such as the sigmoid function. The output produced by the network is then compared with the expected output to calculate the prediction error. The error is propagated backward through the network using the Backpropagation Algorithm, and all weights are updated to reduce the prediction error. This process continues for several epochs until the network converges and achieves satisfactory classification accuracy.

Working of Backpropagation

The Backpropagation Algorithm consists of three major stages. During the **forward pass**, the network computes the output using the current weights. During the **error calculation stage**, the predicted output is compared with the actual output using a loss function such as Mean Squared Error or Binary Cross Entropy. During the **backward pass**, gradients of the error with respect to each weight are calculated

using the chain rule of differentiation. Finally, the weights are updated according to the learning rate so that the prediction error gradually decreases after every iteration.

Confusion Matrix Calculation

The confusion matrix compares the predicted outputs with the actual outputs and summarizes the classification performance of the neural network.

Classification of each example:

Example 1

Actual = Yes

Predicted = Yes

True Positive (TP)

Example 2

Actual = Yes

Predicted = No

False Negative (FN)

Example 3

Actual = No

Predicted = No

True Negative (TN)

Example 4

Actual = Yes

Predicted = Yes

True Positive (TP)

Example 5

Actual = No

Predicted = Yes

False Positive (FP)

The confusion matrix values are therefore

True Positives (TP) = **2**

True Negatives (TN) = **1**

False Positives (FP) = **1**

False Negatives (FN) = **1**

The confusion matrix is

Predicted Yes Predicted No

Actual Yes TP = 2 FN = 1

Actual No FP = 1 TN = 1

Performance Evaluation

1. Accuracy

Accuracy measures the percentage of correctly classified samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

Substitute the values

$$\begin{aligned} &= \frac{2 + 1}{2 + 1 + 1 + 1} \\ &= \frac{3}{5} \\ &= 0.60 \\ Accuracy &= 60\% \end{aligned}$$

2. Precision

Precision measures how many positive predictions are actually correct.

$$\begin{aligned} Precision &= \frac{TP}{TP + FP} \\ &= \frac{2}{2 + 1} \\ &= \frac{2}{3} \\ &= 0.667 \\ Precision &= 66.7\% \end{aligned}$$

3. Recall (Sensitivity)

Recall measures how many actual positive cases are correctly identified.

$$\begin{aligned} Recall &= \frac{TP}{TP + FN} \\ &= \frac{2}{2 + 1} \\ &= \frac{2}{3} \\ &= 0.667 \\ Recall &= 66.7\% \end{aligned}$$

4. F1-Score

The F1-Score is the harmonic mean of Precision and Recall.

$$\begin{aligned} F1 &= \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \\ &= \frac{2 \times 0.667 \times 0.667}{0.667 + 0.667} \\ &= \frac{0.889}{1.334} \\ &= 0.667 \\ F1 \text{ Score} &= 66.7\% \end{aligned}$$

Interpretation of Results

The neural network correctly classified three out of five patient records, resulting in an overall accuracy of **60%**. The Precision value of **66.7%** indicates that approximately two-thirds of the patients predicted to have the disease were actually positive. The Recall value of **66.7%** shows that the model successfully detected two out of the three patients who actually had the disease. The F1-Score of **66.7%** indicates a moderate balance between precision and recall. Although the model demonstrates acceptable performance on this small dataset, its accuracy can be improved by using a larger dataset, additional hidden layers, better feature engineering, and improved hyperparameter tuning.

Role of Data Preprocessing

Data preprocessing plays a crucial role in improving the performance of neural networks. Before training, missing values are handled by replacing them with appropriate statistical values such as the mean or median. Duplicate records are removed to prevent biased learning. Categorical variables such as Blood Pressure, Glucose Level, and Lifestyle Risk are converted into numerical values using encoding techniques. Numerical features are normalized using Min-Max Scaling or Standardization so that all attributes contribute equally during training. Outliers are detected and removed because extreme values can negatively influence weight updates. Proper preprocessing leads to faster convergence, lower prediction error, and better classification accuracy.

Role of Feature Selection

Feature selection identifies the most relevant attributes for disease prediction while eliminating unnecessary or redundant variables. Selecting important features such as Age, Blood Pressure, Glucose Level, and Lifestyle Risk enables the neural network to focus on meaningful information rather than irrelevant data. Techniques such as Correlation Analysis, Information Gain, Chi-Square Test, and Recursive Feature Elimination (RFE) can be used to select informative features. Feature selection reduces computational complexity, minimizes overfitting, improves model interpretability, and increases prediction accuracy on unseen data.

Applications

Neural network classification models are widely used in healthcare for disease diagnosis, cancer detection, diabetes prediction, heart disease analysis, medical image classification, patient risk assessment, drug discovery, and personalized treatment recommendations. Similar classification models are also applied in finance, fraud detection, cybersecurity, speech recognition, sentiment analysis, recommendation systems, and autonomous vehicles.

Advantages

Neural networks automatically learn complex nonlinear relationships from data, achieve high prediction accuracy, continuously improve through training, handle large datasets efficiently, and can generalize well to unseen examples. Their ability to learn hidden patterns makes them one of the most powerful techniques for modern classification problems.

Conclusion

The neural network successfully classifies patient records by learning the relationship between Age, Blood Pressure, Glucose Level, Lifestyle Risk, and disease status using the Backpropagation Algorithm. From the given dataset, the confusion matrix values are **True Positives = 2, True Negatives = 1, False Positives = 1, and False Negatives = 1**. The calculated evaluation metrics are **Accuracy = 60%, Precision = 66.7%, Recall = 66.7%, and F1-Score = 66.7%**. Although the model demonstrates moderate performance on the given dataset, applying effective preprocessing techniques, selecting the most relevant features, increasing the dataset size, and optimizing network parameters can significantly improve the accuracy, reliability, and generalization capability of the neural network.